AI-DRIVEN MAINTENANCE REPORT

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1. Technical Sensor Data

Parameter	Measured Value
Air Temperature	300.0 °F
Process Temperature	340.0 °F
Rotational Speed	1200 RPM
Torque	65.0 Units
Tool Wear	210 %

2. AI Agent Analysis

Agent 1 (Initial Analysis):
Torque and tool wear indicate high risk.

Agent 2 (Verification):
Torque >60 and Wear >200 trigger high risk rules.

3. Engineering Analysis & Recommendations

Summary:
REQUIRED MAINTENANCE

Analysis:
The sensor data indicates a high-risk situation due to abnormal values in torque, tool wear, and temperature. Specifically, the rotational speed is within normal limits, but the air temperature (300.0°C) and process temperature (340.0°C) are significantly higher than expected. This temperature anomaly may be indicative of an underlying issue that requires immediate attention.

Validation:
The sensor data exceeds safety thresholds in three areas:
* Torque: 65.0, exceeding the threshold of >60
* Tool wear: 210, exceeding the threshold of >200
* Temperature difference: (340.0 - 300.0) = 40.0°C, exceeding the threshold of >25°C

Recommendation:
1. IMMEDIATELY reduce rotational speed to minimize further tool wear and potential damage.
2. PERFORM EMERGENCY COOLING PROCEDURES to bring air and process temperatures within acceptable ranges (<275°C).
3. SCHEDULE URGENT TOOL REPLACEMENT or REGRINDING to address excessive wear, as continued operation may lead to catastrophic failure.
Note: These recommendations are based on the provided sensor data and should be reviewed and validated by a qualified maintenance professional before implementation.

4. Conclusion

Final Decision: REQUIRED MAINTENANCE

Decision Source: RULE-BASED SYSTEM TRIGGERED